

IPL Match & Player Performance Analysis

An Exploratory Data Analysis (EDA) project on IPL (Indian Premier League) cricket data, built using Python.

Objective

To analyze IPL match and ball-by-ball data to uncover trends in team performance, toss decisions, and player statistics using data cleaning, aggregation, and visualization.

Dataset

Based on the structure of the **"IPL Complete Dataset (2008–2020)"** (publicly available on Kaggle), containing:

- `matches.csv` — match-level data (teams, toss, venue, winner, season)
- `deliveries.csv` — ball-by-ball data (runs, wickets, batter, bowler)

Dataset source: [Kaggle - IPL Complete Dataset](#)

Tools & Libraries

- **Python**
- **Pandas** — data cleaning and aggregation
- **NumPy** — numerical operations
- **Matplotlib** — data visualization

Analysis Performed

1. **Data Cleaning** — handled missing values, standardized team names across seasons (e.g., renamed franchises)
2. **Season-wise Trends** — matches played per season across 13 seasons (2008–2020)
3. **Team Performance** — top 10 teams by total wins
4. **Toss Analysis** — toss decision trends (bat vs. field) and their relationship with match outcomes
5. **Player Statistics** — top 10 run scorers and top 10 wicket takers across all seasons

Key Findings

- Analyzed 750+ matches across 13 IPL seasons
- Teams increasingly chose to field first after winning the toss, reflecting common T20 chasing strategy

- Winning the toss correlated with winning the match only ~52% of the time — showing limited overall impact
- Identified consistent top performers across seasons in both batting and bowling

Sample Outputs

See the `charts/` folder for all visualizations:

- Matches played per season
- Top 10 winning teams
- Toss decision trend by season
- Top 10 run scorers
- Top 10 wicket takers

How to Run

1. Download `matches.csv` and `deliveries.csv` from the Kaggle link above
2. Place them in the same folder as `ipl_eda.py`
3. Install dependencies:

```
pip install pandas numpy matplotlib
```

4. Run the script:

```
python ipl_eda.py
```

5. Charts will be saved as PNG files in the same folder

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